Original paper

Clinical outcomes of fractionated stereotactic radiotherapy for uveal melanoma – a Polish single-centre experience

Adam Deja[1], Paulina Baluch[2], Marta Wróblewska-Zierhoffer[2], Emilia Zwolińska[2], Maria Radomiak[1], Piotr Milecki[1], Iwona Rospond-Kubiak[2]

[1]The Maria Skłodowska-Curie Greater Poland Cancer Centre, Poznan, Poland
[2]Department of Ophthalmology, Poznan University of Medical Sciences, Poznan, Poland

Introduction: The aim of the study is to present the results of the first Polish cohort of patients treated with fractionated stereotactic radiotherapy (fSRT) for large and/or juxtapapillary uveal melanoma (UM). **Material and methods**: The medical records of the patients who underwent the fSRT for UM in Poznan (cooperation between the Department of Ophthalmology, Poznan University of Medical Sciences and Greater Poland Cancer Centre) in 2014–2020 were retrospectively reviewed. The diagnosis was made in each case after detailed clinical examination, ultrasound and/or choroidal biopsy in selected patients. The irradiation protocol was to deliver 50 Gy in 5 fractions. **Results**: We identified 27 patients, 12 men, 15 women, with a median age of 61.5 years. The median longitudinal basal diameter of treated tumours was 11.7 mm (range 5.2–17.8 mm), median thickness – 7.5 mm (range 1.7–9.7 mm). The median follow-up was 38 months (range 5–101 months). We observed neovascular glaucoma in 9 (34%) patients, which was the reason for globe removal in 3 (11%) of them. There was one local tumour recurrence (3.8%) with extraorbital extension 6 years post-treatment. Local tumour control in our cohort was 96.2%. By the study end, 5 (18.5%) patients died, including 3 (11%) of metastatic disease. The final visual acuity was logMAR 0.3 or better for 3 patients (11%). **Conclusions**: As reported also by other authors, fSRT could be a useful treatment modality for selected large and/or juxtapapillary melanomas.

Key words: uveal melanoma, eye, neoplasm, fractionated stereotactic radiotherapy.

Contemp Oncol (Pozn) 2025; 29 (4): 367–369
DOI: https://doi.org/10.5114/wo.2025.155789

Introduction

Fractionated stereotactic radiotherapy (fSRT) is a non-invasive radiation technique for treating intraocular tumours, notably uveal melanoma. It offers the advantage of tumour targeting without the need for surgical placement of tantalum markers, unlike proton therapy. While fSRT is more widely available than proton beam radiotherapy, it may lack the homogeneous dose distribution of charged particle therapy – a potential disadvantage noted in the literature. Nevertheless, both modalities demonstrate excellent local tumour control and high eye retention rates, supporting their use in selected cases [1, 2].

Common ocular complications after fSRT include cataract formation (up to 67.8%), radiation retinopathy (up to 35.1%), and neovascular glaucoma (NVG), which may occur in up to 20% of cases and is a leading indication for secondary enucleation alongside tumour recurrence. Reported enucleation rates following fSRT range from 3% to16% [1].

In Poland, fSRT has not been widely applied in ocular oncology, and long-term clinical outcomes have not yet been reported.

We aimed to evaluate the safety and efficacy of fSRT in a Polish cohort of patients with uveal melanoma, focusing on tumour control, visual outcomes, treatment-related toxicity, and eye retention.

Material and methods

We conducted a retrospective analysis of all patients with uveal melanoma (UM) treated with fSRT at the Greater Poland Cancer Centre in cooperation with the Department of Ophthalmology, Poznan University of Medical Sciences between 2014 and 2020. All patients treated with fSRT during this time were included. No exclusion criteria were applied beyond standard clinical eligibility for this treatment modality.

Tumours were staged according to the American Joint Committee on Cancer 8th edition classification. Patients were selected for fSRT primarily due to tumour size or location (e.g., juxtapapillary), making them unsuitable for plaque brachytherapy.

Patients were treated using the CyberKnife system, receiving a total dose of 50 Gy in 5 fractions delivered every other day. The fractionation scheme was selected based on prior clinical protocols and literature supporting the balance between efficacy and safety (Table 1).

Immobilization was achieved using a thermoplastic head mask integrated with an arm-like device positioned in front of the eye to limit ocular movement. Real-time eye position was monitored using in-room video cameras connected to the CyberKnife tracking system.

The planning target volume was defined by adding a 2-mm margin to the clinical tumour volume, based on institutional protocols to account for setup uncertainties and minimal eye motion in the absence of implanted

Table 1. Patient characteristics

Parameters	Value
Number of patients	27
Number of female patients	15
Number of male patients	12
Number of patients with hypertension	9
Number of patients with diabetes	1
Mean age	62.04
Median age	61.5
Mean follow-up time (months)	40.65
Median follow-up time (months)	32.0
Mean radiation dose [Gy]	48.14
Median radiation dose [Gy]	50.0
Mean anterior chamber dose – Dmax	35.48
Median anterior chamber dose – Dmax	41.93
Mean anterior chamber dose – Dmean	15.9
Median anterior chamber dose – Dmean	11.25

Table 2. Clinical outcomes

Outcome	Number of cases
Intraorbital recurrences observed in the follow-up period	1
Distant metastases developed after fSRT	5
Radiation-related neuropathy	3
Neovascular glaucoma	9
Globe removal needed	3

fSRT – fractionated stereotactic radiotherapy

markers. Treatment plans were based on non-isocentric, non-coplanar beams, normalized to the 80% isodose line.

Doses to the anterior chamber, lens, optic nerve, and macula were recorded where available. Median and range values were presented.

The first follow-up occurred within 2 months of treatment. Patients were monitored at regular intervals (tyoically every 3–6 months), including assessment of visual acuity, intraocular pressure, slit-lamp biomicroscopy, fundus examination, colour fundus photography, optical coherence tomography (OCT), OCT-A, and ultrasound.

Tumour control was defined as stability or regression in imaging without signs of recurrence. Neovascular glaucoma was diagnosed clinically by anterior segment neovascularization and elevated intraocular pressure. Visual outcomes were recorded at baseline, intermediate timepoints, and final follow-up using logMAR scale.

Adverse events were graded according to computed tomography coronary angiography (CTCAE) v5.0 where applicable. Retinopathy, optic neuropathy, and glaucoma were managed according to standard ophthalmic guidelines (e.g., anti-vascular endothelial growth factor – anti-VEGF, therapy, cryotherapy, pharmacologic agents).

Statistical analysis was performed using Python (SciPy library) [3]. The Mann-Whitney *U* test was used for comparisons between subgroups. Due to the limited sample size, survival analyses were not statistically powered; however, descriptive outcomes (eye preservation, visual function, mortality) are reported. Stratification by follow-up duration (≥ 24 months) was performed to improve consistency (Table 2).

All patients signed informed consent prior to treatment. According to Polish law, no institutional review board approval was required for retrospective chart reviews. The study adhered to the tenets of the Declaration of Helsinki. No patients were lost to follow-up.

Results

We identified 27 patients (12 men, 15 women) with a median age of 62.5 years. The median tumour largest basal diameter (LBD) was 12.35 mm (range 5.22–18.2 mm) and median tumour thickness was 7.66 mm (range 1.7–9.7 mm). Median follow-up was 38 months (range 5–101 months); 20 patients (74%) were followed ≥ 24 months.

All patients received a median total dose of 50 Gy. The median maximum dose to the anterior chamber was 42 Gy, and the mean dose was 11.3 Gy. Where available, the median maximum doses to other critical structures were: optic nerve: 31.6 Gy (range 19.2–47.8 Gy); lens: 9.4 Gy (calculated 2.8–17.5 Gy); fovea: 24.2 Gy (range 12.1–39.6 Gy).

Local tumour control was achieved in 26 of 27 patients (96.3%). One patient (3.7%) experienced tumour recurrence 6 years post-treatment involving a juxtapapillary melanoma with extraorbital extension. This patient underwent extended enucleation.

At final follow-up, 3 patients (11%) retained visual acuity of logMAR 0.3 or better (defined as functional vision); 4 patients (15%) had visual acuity of logMAR ≤ 1.0; 20 patients (74%) had moderate to severe vision loss (logMAR > 1.0); and 14 patients (52%) hadno light perception.

Neovascular glaucoma occurred in 10 patients (37%), median onset at 20 months (range 5–35 months). Computed tomography coronary angiography grades: grade 2 in 4 patients (controlled medically), grade 3 in 3 (requiring anti-VEGF or cryotherapy), and grade 4 in 3 patients (leading to enucleation). Neovascular glaucoma was significantly associated with tumour LBD and thickness ($p < 0.05$).

Radiation retinopathy was diagnosed in 6 patients (22%) with median onset at 18 months (Grade 1–2: 4 patients; Grade 3: 2 patients (requiring intravitreal injections)). Optic neuropathy occurred in 2 patients (7.4%), both with juxtapapillary tumour location (Grade 2–3).

Enucleation was required in 3 patients (11.1%): 2 due to uncontrolled painful NVG (grade 4), and 1 due to tumour recurrence.

By the study end, 5 patients (18.5%) had died, of which 3 (11.1%) due to metastatic uveal melanoma. No patients were lost to follow-up.

In patients followed ≥ 24 months ($n = 20$): NVG rate was 40%; visual acuity logMAR ≤ 0.3: 11%; local tumour control: 95%; enucleation: 3 cases (15%).

Discussion

This study presents the first reported outcomes of fSRT for UM in Poland, with long-term follow-up. Fractionated stereotactic radiotherapy was used in patients unsuitable for ruthenium plaque brachytherapy due to tumour size or juxtapapillary location. Our findings support the feasibility and efficacy of this modality, with local control of 96.3%, an eye retention rate of 88.9%, and functional vision (logMAR ≤ 0.3) preserved in 11% of patients at final follow-up.

The observed eye retention rate is comparable to those reported in other fSRT series, with 83.3–90.7% [1, 2]. Similarly, our local control rate aligns with large retrospective cohorts using fSRT or stereotactic radiosurgery, including CyberKnife and GammaKnife platforms.

Neovascular glaucoma, a major late complication, occurred in 37% of our patients – higher than in some series reporting 19–27% [4], though consistent with our cohort's inclusion of larger tumours and juxtapapillary lesions. Tumour size was significantly associated with NVG, confirming previous findings [5, 6]. Other known risk factors, including ciliary body involvement and higher radiation dose to anterior segment structures, could not be fully analysed due to sample size limitations but are relevant for future studies.

Radiation retinopathy and optic neuropathy were less frequent (22% and 7.4%, respectively) and occurred later during follow-up. Visual acuity outcomes were heterogeneous, reflecting tumour location, initial vision, and radiation-related toxicities. Notably, 11% of patients retained functional vision, suggesting that fSRT remains a vision-preserving option in selected cases.

Comparison with other modalities highlights the nuanced role of fSRT.

Proton beam therapy, though dosimetrically advantageous, does not demonstrate superior local control compared to fSRT. A comparative study by van Beek *et al*. showed no significant difference in tumour control or eye retention, though vitreous haemorrhages were less common with proton therapy [7]. Plaque brachytherapy offers excellent eye retention but may be limited in juxtapapillary tumours or those exceeding 12 mm in diameter [8]. In Sagoo *et al*. study on juxtapapillary melanomas, local control was lower than in our fSRT cohort [9].

Stereotactic radiosurgery delivers single-fraction high-dose therapy and has shown comparable local control to fSRT in meta-analyses [6], but may carry a higher risk of early toxicities.

The decision to use fSRT in our centre was influenced by availability of the CyberKnife system, patient-specific anatomy.

Although the absence of randomized comparisons limits definitive conclusions, current evidence suggests that fSRT provides an effective alternative to other radiotherapy modalities, particularly where plaque or proton therapy is not feasible.

Emerging approaches – such as prophylactic anti-VEGF therapy – may further improve visual outcomes and reduce radiation retinopathy in the future [10].

This retrospective study has limitations, including small sample size, absence of a control group, and variability in follow-up duration. Not all imaging and dosimetric data were consistently available. Visual acuity reporting was limited by documentation practices, and CTCAE grading was applied retrospectively.

Nevertheless, the real-world data provide insight into the clinical performance of fSRT in a Central European context and underscore its value in managing complex UM cases.

Conclusions

Fractionated stereotactic radiotherapy appears to be a safe and effective modality for selected patients with uveal melanoma, achieving high local control and acceptable toxicity. Further prospective studies with standardized protocols are needed to validate these findings.

Disclosures

1. Institutional review board statement: Not applicable.
2. Assistance with the article: None.
3. Financial support and sponsorship: None.
4. Conflicts of interest: None.

References

1. Van Beek JGM, van Rij CM, Baart SJ, Yavuzyigitoglu S, Bergmann MJ, Paridaens D, et al. Fractionated stereotactic radiotherapy for uveal melanoma: long-term outcome and control rates. Acta Ophthalmol 2022; 100: 511-519.
2. Van Beek JGM, Buitendijk GHS, Timman R, Muller K, Luyten GPM, Paridaens D, et al. Quality of life: fractionated stereotactic radiotherapy versus enucleation treatment in uveal melanoma patients. Acta Ophthalmol 2018; 96: 841-848.
3. Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nat Methods 2020; 17: 261-272
4. Özcan G, Gündüz AK, Mirzayev İ, Oysul K, Uysal H. Early results of stereotactic radiosurgery in uveal melanoma and risk factors for radiation retinopathy. Turk J Ophthalmol 2020; 50: 156-162.
5. Hirasawa N, Tsuji H, Ishikawa H, Koyama-Ito H, Kamada T, Mizoe JE, et al. Risk factors for neovascular glaucoma after carbon ion radiotherapy of choroidal melanoma using dose-volume histogram analysis. Int J Radiat Oncol Biol Phys 2007; 67: 538-543.
6. Kosydar S, Robertson JC, Woodfin M, Mayr NA, Sahgal A, Timmerman RD, et al. Systematic review and meta-analysis on the use of photon-based stereotactic radiosurgery versus fractionated stereotactic radiotherapy for the treatment of uveal melanoma. Am J Clin Oncol 2021; 44: 32-42.
7. Van Beek JGM, Jackelien GM, van Rij CM, Baart SJ, Paridaens D, van den Bosch WA, et al. Local tumour control and radiation side effects for fractionated stereotactic photon beam radiotherapy compared to proton beam radiotherapy in uveal melanoma. Radiother Oncol 2021; 157: 219-224.
8. Laliscia C, Perrone F, Guido F, Cresti F, Fuentes T, Paiar F et al. A cohort of long-surviving patients affected by small and medium uveal melanoma treated with ruthenium-106 plaque brachytherapy. Contemp Oncol 2024; 28: 318-325.
9. Sagoo MS, Shields CL, Emrich J, Mashayekhi A, Komarnicky L, Shields JA. Plaque radiotherapy for juxtapapillary choroidal melanoma: treatment complications and visual outcomes in 650 consecutive cases. JAMA Ophthalmol 2014; 132: 697-702.
10. Finger PT, Chin KJ, Semenova EA. Intravitreal anti-VEGF therapy for macular radiation retinopathy: a 10-year study. Eur J Ophthalmol 2016; 26: 60-66.

Address for correspondence

Iwona Rospond-Kubiak, PhD
Department of Ophthalmology
Poznan University of Medical Sciences
Poznan, Poland
e-mail: irospondkubiak@ump.edu.pl

Submitted: 30.06.2025
Accepted: 02.09.2025